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Piezoelectric Properties of Triglycine-Sulphate

Takuro IKEDA, Yoichi TANAKA and Hiroo TOYODA

*Electrical Communication Laboratory, Nippon Telegraph and
Telephone Public Corporation, Musashino, Tokyo*

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Since triglycine sulphate crystal is centrosymmetric above the Curie point, the piezoelectricity in the ferroelectric phase is expected to be an "induced" one as in the case of barium titanate, distinct from an "intrinsic" piezoelectricity of rochelle salt. Temperature dependences of piezoelectric constants are measured by using various cuts of crystal, and some experiments on electrostrictive effect above the Curie point are performed as well. We then obtain electrostrictive constants, and the above expectation is confirmed in several points. Moreover, the phase transition is found not to be accompanied by any elastic change. Next, some effects of irradiation by x-ray and γ -ray on the piezoelectric constants are examined, and it will be shown that an irradiation is useful for preventing a diminution of piezoelectricity resulting from growth of antiparallel domains, especially near the Curie point.

§ 1. Introduction

Since the ferroelectricity in triglycine-sulphate $(C_2NO_2H_3)_3 \cdot H_2SO_4$ was discovered in 1956, many reports have been published on its interesting properties.

The paraelectric triglycine sulphate be-

longs to the monoclinic prismatic class, the space group being $C_{2h}^2 - P2_1/m$, and shows no piezoelectricity. At $47.5^\circ C$, it is transformed to the ferroelectric phase, and the space group changes to $C_2^2 - P_2$, which belongs to the polar spheroidal class and is piezoelectric.

The lattice parameters have been measured by Wood¹⁾, and the atomic positions have been determined by Hoshino *et al.*²⁾. The temperature dependences of lattice parameters have been measured by Ezhkova *et al.*³⁾. The transition from a ferro- to a paraelectric state is believed to be of the second order, and of the order-disorder type. The ferroelectric axis is parallel to b-axis and the polarization reversal is characterized by a transformation from a right-handed system to a left-handed one.

The piezoelectric properties of triglycine-sulphate have been reported by a few authors. Konstantinova and others⁴⁾ obtained all of the dielectric, elastic and piezoelectric constants at room temperature by using various cuts of crystal. Husimi and Kataoka⁵⁾ also reported the piezoelectric constants, except shear components. The values reported in these two papers, however, differ considerably from each other; for instance, the value of d_{23} obtained by the latter authors are three times as large as that by the former. It is clear that this discrepancy is due to polydomain structure, and so the previous experiment performed without prepoling may be unreliable. The temperature dependences of the piezoelectric constants have rarely been obtained, though Husimi showed a datum on d_{23} constant in an arbitrary unit. Recently, Gilletta⁶⁾ also measured the elastic and piezoelectric constants. Any detailed discussions on the piezoelectricity have never been given.

The crystal of barium titanate shows no piezoelectricity above the Curie point, but in the ferroelectric state it is generally accepted that the spontaneous polarization gives rise to an "induced" piezoelectricity by the effect of electrostriction as explained by Devonshire⁷⁾. This situation is sharply distinct from the piezoelectricity of rochelle salt, which is an "intrinsic" piezoelectric even above the Curie point. The present case, in which no "intrinsic" piezoelectricity exists above the Curie point, is similar rather to the case of barium titanate than rochelle salt, and the observed piezoelectricity appears to be "induced".

In the present study, temperature dependences of all piezoelectric constants are measured, and some data on electrostrictive effect above the Curie point are given as well.

The electrostrictive constants determined from the present experiment are found to be independent of temperature, and this fact will give an evidence of an "induced" piezoelectricity. Sil'vestrova⁸⁾ reported some electrostrictive data, including effects of field on the electromechanical coupling coefficient and elastic compliance coefficient, but she gave no further discussions.

Some effects of the radiation on the properties of the crystal have been already examined⁹⁾, and it is known that the irradiation by x-ray and neutron suppresses a reversal of polarization. It may be expected that the irradiation during prepoling quenches a single domain structure and gives a maximum piezoelectricity. The irradiation effects on the piezoelectricity will be also studied in the present paper. A part of the present results was already published¹⁰⁾.

§ 2. Piezoelectric and Electrostrictive Relations

Fundamental piezoelectric relations, combining the electric and elastic properties, are described in various ways, according to the choice of independent variables. For instance, they are expressed as follows:

$$\left. \begin{aligned} S_i &= s_{ij}^F T_j + d_{ni} E_n, \\ P_n &= k_{mn}^F E_m + d_{mj} T_j, \end{aligned} \right\} \quad (2.1)$$

and

$$\left. \begin{aligned} S_i &= s_{ij}^F T_j + b_{ni} P_n, \\ E_m &= \chi_{mn}^F P_n - b_{mj} T_j, \end{aligned} \right\} \quad (2.2)$$

where the notations have the following meanings;

- S : strain,
- T : stress,
- P : polarization,
- E : field,
- s : elastic compliance coefficient,
- κ : dielectric susceptibility,
- χ : reciprocal susceptibility,
- d and b : piezoelectric constant,
- suffixes E and P : constant field and constant polarization,
- suffix F : stress free.

In the case of ferroelectric triglycine-sulphate, some components of s , κ and d must vanish, on account of the crystal symmetry C_2 . Namely, the matrices (s_{ij}) , (κ_{mn}) and (d_{mj}) are the following ones:

$$\left\{ \begin{array}{l} \begin{bmatrix} S_{11} & S_{12} & S_{13} & 0 & S_{15} & 0 \\ S_{12} & S_{22} & S_{23} & 0 & S_{25} & 0 \\ S_{13} & S_{23} & S_{33} & 0 & S_{35} & 0 \\ 0 & 0 & 0 & S_{44} & 0 & S_{46} \\ S_{15} & S_{25} & S_{35} & 0 & S_{55} & 0 \\ 0 & 0 & 0 & S_{46} & 0 & S_{66} \end{bmatrix}, \begin{bmatrix} \kappa_{11} & 0 & \kappa_{13} \\ 0 & \kappa_{22} & 0 \\ \kappa_{13} & 0 & \kappa_{33} \end{bmatrix} \\ \text{and} \\ \begin{bmatrix} 0 & 0 & 0 & d_{14} & 0 & d_{16} \\ d_{21} & d_{22} & d_{23} & 0 & d_{25} & 0 \\ 0 & 0 & 0 & d_{34} & 0 & d_{36} \end{bmatrix} \end{array} \right\} \quad (2.3)$$

The matrices (χ_{mn}) and (b_{mj}) have the same forms as (κ_{mn}) and (d_{mj}) , respectively.

In the paraelectric phase, the symmetry changes to C_{2h} and then all components of d and b matrices are zero. But, when a d.c. biasing voltage is applied to the crystal, an apparent piezoelectricity is observed to occur, as shown in § 4(b), and therefore, an electrostriction must be taken into account.

The electrostrictive relations are introduced just as in the case of barium titanate developed by Devonshire⁷⁾. Namely, the corresponding equations above the Curie point are expressed in the following tensor forms:

$$\left\{ \begin{array}{l} S_{ij} = s_{ijkl}^P T_{kl} + Q_{ijn0} P_n P_0, \\ \text{and} \\ E_m = \chi_{mn}^F P_n - 2Q_{ijmn} T_{ij} P_n, \end{array} \right\} \quad (2.4)$$

where Q_{ijn0} is the electrostrictive constant and given by the equation

$$2Q_{ijn0} = \frac{\partial^2 S_{ij}}{\partial P_n \partial P_0} = - \frac{\partial^2 E_0}{\partial T_{ij} \partial P_n}.$$

It will be easily shown that this electrostrictive effect causes an "induced" piezoelectricity below the Curie point, as described in § 1. Namely, the spontaneous polarization P_s occurs in parallel with y -axis in the ferroelectric phase, and the polarization P_2 in eqs. (2.4) is replaced by $P_s + P_2$. When constant terms and higher order ones are neglected, eqs. (2.4) can be transformed to the same as (2.2) or (2.1). Then, the following equations among various constants are obtained:

$$\left\{ \begin{array}{l} \kappa_{11}^F = \chi_{33}^F / D, \quad \kappa_{22}^F = 1 / \chi_{22}^F, \\ \kappa_{33}^F = \chi_{11}^F / D, \quad \kappa_{13}^F = -\chi_{13}^F / D, \\ (D = \chi_{11}^F \chi_{33}^F - \chi_{13}^{F2}) \end{array} \right\} \quad (2.5)$$

$$\left\{ \begin{array}{l} s_{ij}^B = s_{ij}^P + b_{2i} d_{2j} \quad (i, j = 1, 2, \dots, 6), \\ s_{44}^B = s_{44}^P + b_{14} d_{14} + b_{34} d_{34}, \\ s_{66}^B = s_{66}^P + b_{16} d_{16} + b_{36} d_{36}, \\ s_{46}^B = s_{46}^P + b_{16} d_{14} + b_{36} d_{34}, \end{array} \right\} \quad (2.6)$$

$$\left\{ \begin{array}{l} d_{2i} = b_{2i} \kappa_{22}^F \quad (i = 1, 2, 3, 5), \\ d_{14} = b_{14} \kappa_{11}^F + b_{34} \kappa_{13}^F, \\ d_{34} = b_{14} \kappa_{13}^F + b_{34} \kappa_{33}^F, \\ d_{16} = b_{16} \kappa_{11}^F + b_{36} \kappa_{13}^F, \\ d_{36} = b_{16} \kappa_{13}^F + b_{36} \kappa_{33}^F, \\ b_{2i} = 2Q_{2i} P_s \quad (i = 1, 2, 3, 5), \\ b_{14} = b_{36} = Q_{46} P_s, \\ b_{34} = Q_{44} P_s, \\ b_{16} = Q_{66} P_s. \end{array} \right\} \quad (2.7)$$

$$\left\{ \begin{array}{l} b_{2i} = 2Q_{2i} P_s \quad (i = 1, 2, 3, 5), \\ b_{14} = b_{36} = Q_{46} P_s, \\ b_{34} = Q_{44} P_s, \\ b_{16} = Q_{66} P_s. \end{array} \right\} \quad (2.8)$$

In eqs. (2.8), Q_{ij} 's are used instead of tensor components Q_{ijmn} 's where $Q_{iijj} = Q_{ij}$, $2Q_{iimn} = Q_{is}$ and $4Q_{ijmn} = Q_{rs}$ ($i, j, m, n = 1, 2, 3; r, s = 4, 5, 6; 23 \rightarrow 4, 31 \rightarrow 5, 12 \rightarrow 6$). This simplification of suffices is the same as that between s_{ij} and s_{ijkl} . The matrix of (Q_{ij}) is the same as that of (s_{ij}) , but only the components related to P_2 appear in the above equations.

The relations described above suggest some means to confirm an expectation on an "induced" piezoelectricity in triglycine sulphate crystal. The electrostrictive constants Q_{ij} 's are determined at various temperatures below the Curie point from piezoelectric data by using (2.7) and (2.8) on one hand, and from electrostrictive experiments above the Curie point on the other hand. If the constants thus obtained are nearly independent of temperature, the expectation will be accepted.

§ 3. Experimental Procedures

All specimens were cut from crystals grown from an aqueous squalous solution. Shape of a crystal as grown is schematically shown in Fig. 1. The piezoelectric constants are related to the rectangular coordinates xyz instead of the monoclinic axes abc as shown

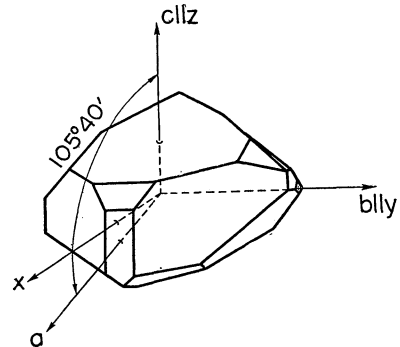


Fig. 1. Coordinates and Crystallographic Axes.

in Fig. 1.

The piezoelectric and elastic measurements were carried out by the self-resonance method using bar-shaped specimens. The orientations of specimens cut are shown in Fig. 2(a), (b) and (c). In the present paper, specimens are named as shown there.

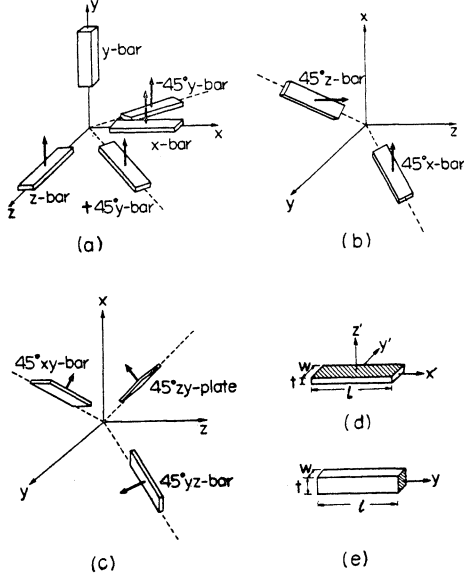


Fig. 2. Orientations of Specimens.

In the measurements of bars except y-bar, let us take rotated axes $x'y'z'$ for convenience as shown in Fig. 2(d), where x' and z' are parallel to length and thickness directions respectively. On both faces perpendicular to z' axis were evaporated gold electrodes, and a mechanical vibration was generated in x' direction by applying an electric field parallel to z' . Short arrows in (a), (b) and (c) indicate their z' directions.

When a resonant and an antiresonant frequencies are denoted by f_R and f_A and an electric capacity measured at frequency far below f_R is by C^F , some physical constants are determined by the following equations:

$$\left. \begin{aligned} \frac{k_{31}^2}{1-k_{31}^2} &= \frac{\pi^2}{4} \frac{f_A - f_R}{f_R}, \\ k_{31}^2 &= \frac{4\pi d_{31}^2}{s_{11}^{E'} \epsilon_{33}^{F'}}, \\ f_R &= \frac{1}{2l} \sqrt{\frac{1}{\rho s_{11}^{E'}}}, \\ \text{and} \\ C^F &= \frac{\epsilon_{33}^{F'} lw}{4\pi t}, \end{aligned} \right\} \quad (3.1)$$

where k_{31}' is an electromechanical coupling coefficient, and d_{31}' , $s_{11}^{E'}$ and $\epsilon_{33}^{F'}$ are a piezoelectric constant, an elastic compliance and a dielectric constant as explained in § 2. Primes mean that they are related to the $x'y'z'$ coordinates. ρ is the density ($=1.68$), and l , w and t are dimensions of a bar as shown in the figure.

These constants measured are easily related to the constants referring to the xyz system (see Mason's book⁽¹¹⁾), and the relations are listed in Table I. The $45^\circ zx$ plate was used only to determine ϵ_{13} .

Only in the y-bar, electrodes were attached to end faces as shown in Fig. 2(e), where both electric field and mechanical strain are parallel to y . In this case, the equations corresponding to (3.1) are replaced by the following ones⁽¹²⁾:

$$\left. \begin{aligned} k_{22}^2 &= \frac{\pi^2}{4} \frac{f_A - f_R}{f_A}, \\ k_{22}^2 &= \frac{4\pi d_{22}^2}{\epsilon_{22}^{E'} s_{22}^{E'}}, \\ f_A &= \frac{1}{2l} \sqrt{\frac{1}{\rho s_{22}^{E'}}}, \\ s_{22}^D &= s_{22}^{E'} (1 - k_{22}^2), \\ \text{and} \\ C^F &= \frac{\epsilon_{22}^{F'} wt}{4\pi l}. \end{aligned} \right\} \quad (3.2)$$

Above relations are applicable to a single domain crystal. The crystal usually obtained has complicated domain structures, and then a prepoling by a d.c. field is necessary to get a single domain. But even a prepoling cannot always keep a single domain, especially near the Curie point. Hence, most of the measurements were performed by applying d.c. bias of 90 V/cm c.a., after prepoling under a higher field of about 900 V/cm. In the case of $45^\circ x$ - or $45^\circ z$ -bar in Fig. 2(b), however, spontaneous polarization axis y is parallel to the electrode face, and then, any effective biasing field cannot be applied during the measurement. An irradiation fixes a single domain, as shown below in the case of y-cut specimens. The crystal, therefore, was temporarily electroded on the transverse $x'z'$ faces and irradiated by x-ray for one hour under a high d.c. field. Then, these electrodes were removed and new electrodes were attached to the $x'y'$

faces. The measurements were done without any bias.

In determining signs of the piezoelectric constants, we must consider here on the sense of y axis, which is reversible under an external field. It makes the matter more complicated that a polarization reversal in this crystal is a transformation from d-crystal to l-crystal or vice versa. The signs of d_{14} , d_{25} and d_{36} depend upon the direction of polarization together with the choice of the coordinate system xyz . Hence we shall define the coordinate system here as follows: When the crystal is prepoled under a high field, the positive sense of y -axis is specified to be the same as that of the prepoling field, and other axes x and z are related to a and c as shown in Fig. 1. (In this definition, however, the left-handed system is also involved, and in this case, if we use the right-handed system, d_{14} , d_{25} and d_{36} change their signs.)

The capacity of specimen is often small and an influence of a stray capacity is unable to be neglected. When a clamped capacity and a stray one are denoted by C_0 and C_s respectively, the observed antiresonant frequency f_A must be corrected by an equation $f_A^2 - f_R^2 = (C_0 + C_s)/C_0 \cdot (f_A'^2 - f_R^2)$ to obtain the true f_A , where C_0 is equal to $(1 - k^2)C^F$, but the correction was made by using C^F instead of C_0 .

The elastic compliance coefficient s^D for

constant electric displacement can be determined by a relation $s^D = s^B(1 - k^2)$, but this means is not always reliable. In the present study, the values of s_{22}^D , s_{11}^D and s_{33}^D were obtained also by the composite-bar method. The bar of bare crystal with its length parallel to x , y or z axis was cemented to a quartz vibrator endwise. The specimens were prepoled in y direction before cemented. It is known that the resonant frequency of bare specimen calculated from an observed frequency of the composite-bar gives s^D ¹³⁾.

The bars with required orientations were cut by using natural faces as grown as clues. The true orientations were examined by an x-ray precession camera. Errors of orientations were $\pm 1^\circ$ or so in most of the y -cut specimens in Fig. 2(a). In determining shear components d_{14} , d_{36} , d_{16} and d_{34} , effects of slight misorientations cannot be ignored because of large values of d_{24} 's, especially at higher temperature near the Curie point. Hence, the equations for d'_{31} in Table I were changed to those containing actual direction cosines for calculating shear components, and are shown in footnotes of the table.

Some effects of irradiation by x -ray or γ -ray on the piezoelectric constant were examined. The specimen of z -bar was laid at distance of 55 mm from an x-ray tube with copper anticathode and irradiated by x-ray (35 kV, 15 mA) under biasing field of 900

Table I

	z'	x'	ϵ'	d'	s'
x -bar	y	x	ϵ_{22}	d_{21}	s_{11}
z -bar	y	z	ϵ_{22}	d_{23}	s_{33}
$\pm 45^\circ y$ -bar	y	45° to x and $\pm z$	ϵ_{22}	$\frac{1}{2}(d_{21} + d_{23} \pm d_{25})$	—
$45^\circ x$ -bar	x	45° to y and z	ϵ_{11}	$\frac{1}{2}d_{14}^{*1)}$	$\frac{1}{4}(s_{22} + s_{33} + s_{44} + 2s_{23})$
$45^\circ z$ -bar	z	45° to x and y	ϵ_{33}	$\frac{1}{2}d_{36}^{*2)}$	$\frac{1}{4}(s_{11} + s_{22} + s_{66} + 2s_{12})$
$45^\circ yz$ -bar	45° to x and $-y$	45° to x and y	$\frac{1}{2}(\epsilon_{11} + \epsilon_{22})$	$-\frac{1}{2\sqrt{2}}(d_{21} + d_{22} - d_{16}^{*3)})$	—
$45^\circ yz$ -bar	45° to y and $-z$	45° to y and z	$\frac{1}{2}(\epsilon_{22} + \epsilon_{33})$	$\frac{1}{2\sqrt{2}}(d_{22} + d_{23} - d_{34}^{*4)})$	—
$45^\circ zx$ -plate	45° to x and $-z$	45° to z and x	$\frac{1}{2}(\epsilon_{11} + \epsilon_{33} + 2\epsilon_{13})$	—	—
y -bar			ϵ_{22}	d_{22}	s_{22}

* Taking account of actual orientations,

1) $-0.00585d_{22} - 0.00613d_{23} + 0.498d_{14}$.

2) $0.0147d_{21} + 0.0115d_{22} + 0.496d_{36}$.

3) $-0.353d_{21} - 0.353d_{22} - 0.0028d_{23} + 0.010d_{25} + 0.353d_{16}$.

3) $0.322d_{22} + 0.391d_{23} - 0.333d_{34}$.

V/cm at first for half an hour. The temperature dependence of d_{23} was measured without bias. It was irradiated again for one hour and the measurement was repeated. At last, it was done after irradiation for three hours more. Moreover, aging of many irradiated specimens were observed for 75 days at 42°C.

Similar experiments were carried out in the case of γ -irradiation. The source of Co^{60} (2000 Curie) in the Tokai Laboratory of our institute was used; the intensity of γ -ray was about $1.8 \cdot 10^5$ Roentg./hr, and time is taken as parameter up to 6 hrs.

§ 4. Results and Discussions

(a) Y-cut Specimens

Elastic compliance coefficient of the z-bar is plotted against temperature in Fig. 3(a). Value of s_{33}^E increases towards the Curie point and then drops abruptly to a nearly constant values. That of s_{33}^D calculated from s_{33}^E is shown by a broken line, which shows a narrow peak only near the Curie point.

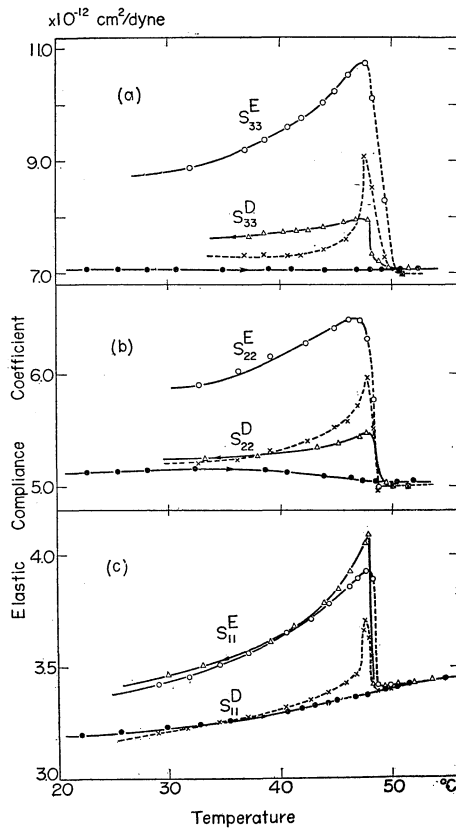


Fig. 3. Elastic Compliance Coefficients.

The composite-bar method, however, gives a different result; namely, the value indicated by a full curve in a heating run is nearly independent of temperature, but in a cooling run, it suddenly rises at the Curie point and decreases gradually. These facts may be understood as described below.

The compliance s_{33}^D of a bare z-bar prepoled is independent of temperature, as shown by a heating full curve. When the crystal is heated above the Curie point and then cooled down below it, many antiparallel domains occur and bring a value of compliance closer to s_{33}^E apart from s_{33}^D ; therefore, a cooling curves is intermediate between curves of s_{33}^E and s_{33}^D . In the self-resonance method, a determination of s_{33}^D is closely related to the piezoelectricity, on the contrary to the composite-bar method. If temperature is not uniform throughout the crystal, it may occur in a temperature range near the Curie point that one part is below and other part is above the Curie point. If it is the case, an apparent diminution of piezoelectricity will cause an increase of s_{33}^D , as shown by a broken line.

Similar phenomena are found in the cases of y-bar and x-bar. The results for them are shown in Fig. 3(b) and (c). Although s_{22}^D decreases and s_{11}^D increases slightly with increasing temperature, both of them show no anomalies in the range near the Curie point. The elastic compliance coefficient s^D is little different from s^E , and it may be considered to be an intrinsic elastic parameter. Accordingly, it will be concluded that the

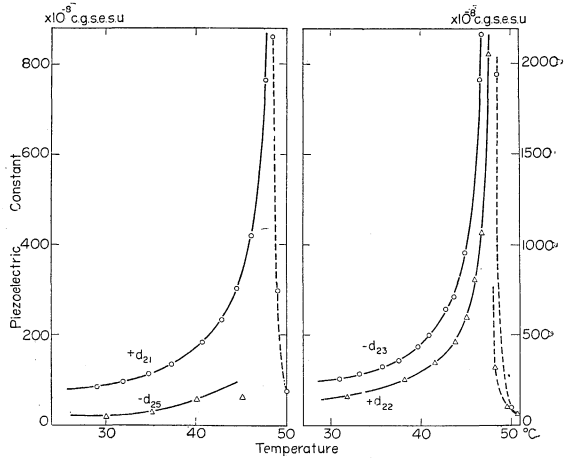


Fig. 4. Piezoelectric Constants d_{2i} 's.

transition at the Curie point is not accompanied by any elastic change.

The piezoelectric constants d_{21} , d_{22} and d_{23} were determined by using x -, y - and z -bars, and shown in Fig. 4. The value of d_{25} was obtained by $\pm 45^\circ y$ -bars, and shown also in the figure. All of the curves rise sharply towards the Curie point. Apparent piezoelectricity above the Curie point, shown by dotted lines, is a result of electrostriction under the biasing field, as seen below.

(b) *Electrostriction*

Above the Curie point, a large electrostrictive effect is seen. When a d.c. field is applied on the z -bar, self-resonant vibration is observed. Apparent piezoelectric constant d_{23} is measured at various temperatures under a few stages of d.c. biasing fields, and the

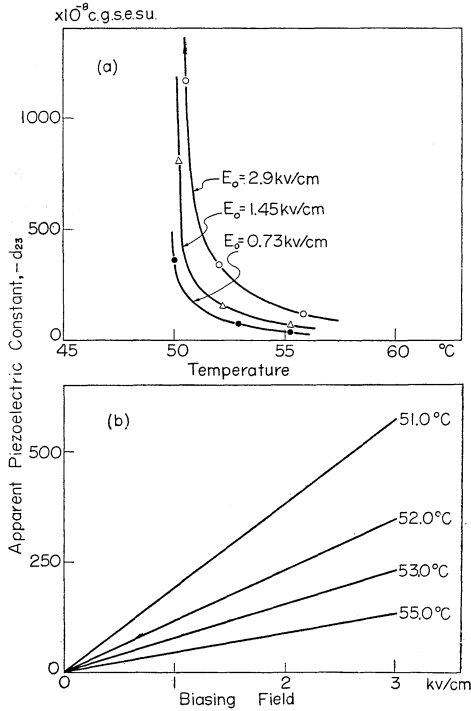


Fig. 5. Electrostriction of the z -bar.

results are shown in Fig. 5(a). When d_{23} is plotted against biasing field at various temperatures, it is found to be proportional to the bias. The electrostrictive constant Q_{23} is able to be obtained in the same way as in the case of "induced" piezoelectricity described in § 2; namely, when the d.c. polarization P_0 is used instead of P_s and the dielectric susceptibility is assumed

to be independent of a magnitude of electric field, then a relation $d_{23} = 2Q_{23}\kappa_{22}^F P_0 = 2Q_{23}\kappa_{22}^F E_0$ is obtained, where E_0 is the biasing field. The values of Q_{23} thus determined are plotted above the Curie point in Fig. 6.

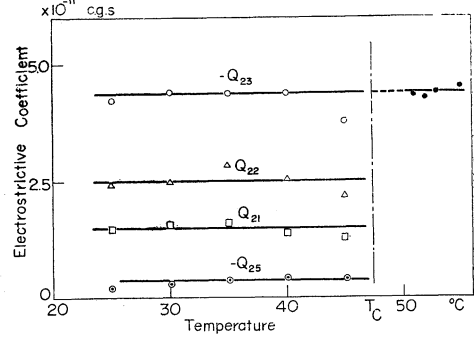


Fig. 6. Electrostrictive Constants Q_{2i} 's.

The values of Q_{2i} ($i=1, 2, 3$ and 5), on the other hand, are given by using piezoelectric constants d_{2i} below the Curie point, as described in § 2. The results are shown also in Fig. 6. It will be easily seen that the electrostrictive constants Q_{2i} thus obtained are nearly independent of temperature and therefore the concept of an "induced" piezoelectricity is confirmed in a part. The electrostrictive constants thus determined are as follows:

$$Q_{21}=1.5, \quad Q_{22}=2.5, \quad Q_{23}=-4.4 \text{ and } Q_{25}=-0.4 \\ \text{in } 10^{-11} \text{ c.g.s. units.}$$

(c) *Shear Components*

The results on the $45^\circ x$ - and $45^\circ z$ -bars are shown in Fig. 7. The values of d_{14} and d_{36} decrease gradually with increasing temperature. Other constants are nearly independent of temperature, and slight increases seem to be due to errors of orientations in cutting, but effects of misorientations are to be smaller in them.

On the other hand, it is more difficult to determine the piezoelectric constants d_{16} and d_{34} ; for the values of d_{31}' in the $45^\circ xy$ - and $45^\circ yz$ -bars include d_{22} , d_{21} or d_{23} , which are large and dependent considerably upon temperature. The results are shown in Fig. 8, where the dielectric constant ϵ_{13} is also shown. A peak observed in ϵ_{13} may be due to misorientation and ϵ_{13} will be considered to be independent of temperature, as in the cases of ϵ_{11} and ϵ_{33} . The values of d_{16} and d_{34} are seen to decrease with increasing temperature,

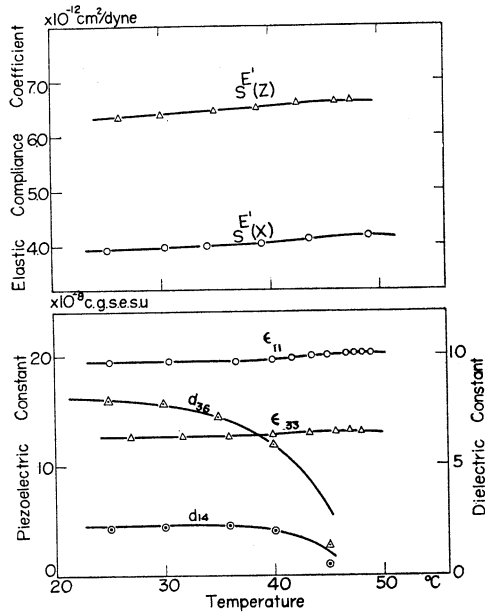
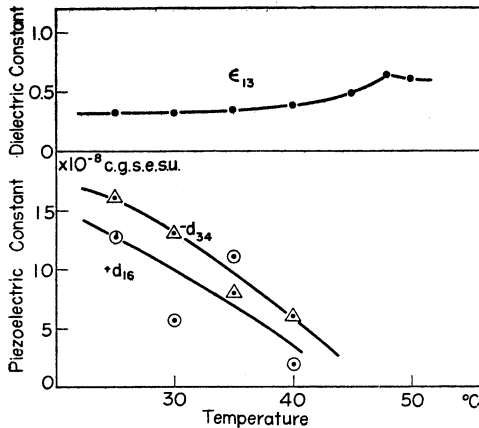


Fig. 7. Data for the 45°x- and 45°z-bars.

Fig. 8. Data of d_{16} , d_{34} and ϵ_{13} .

though slight errors in determining d_{2i} 's affect the results more seriously in these cases than in the case of 45°x- and 45°z-bars.

The electrostrictive theory requires that such constants as d_{14} , d_{34} , d_{16} and d_{36} decrease with temperature in similar behaviours to P_s , as shown in eqs. (2.7) and (2.8). Experimental results seem to show such tendencies to some extent.

The values of shear components at room temperature seem to be rather reliable than those at higher temperature, for influences of d_{2i} 's become considerable in the latter case. Accordingly, at 25°C, let us evaluate b_{14} , b_{36} , b_{16} and b_{34} by using experimental values of d -constants; then, we obtain $b_{14}=5$,

$b_{36}=35$, $b_{16}=18$ and $b_{34}=-35$ in 10^{-8} c.g.s. units. Above results seem to be inconsistent with the electrostrictive theory which requires an equality of b_{14} and b_{36} .

Nonetheless, taking account of difficulties in determining shear components, it will not be asserted that the present results on shear components are insufficient to confirm our expectation of an "induced" piezoelectricity in triglycine sulphate crystals. In the 45° x bar, a depolarization may have occurred in evaporating the second electrodes, owing to warming up.

When the electrostrictive constants other than Q_{2i} 's are calculated by (2.8), we obtain $Q_{44}=-4.3$, $Q_{66}=2.2$ and $Q_{46}=4.2$

in 10^{-11} c.g.s. units, where the value of Q_{46} is now determined from b_{36} .

(d) Effects of Irradiation

Some effects of irradiation by x-ray and γ -ray were examined as described in §3. The piezoelectric constant d_{23} at various temperatures is shown in Fig. 9(a) against time of x-ray irradiation. Though it is increased by irradiation, the reason is not understood. Remarkable effects are seen in the resistance

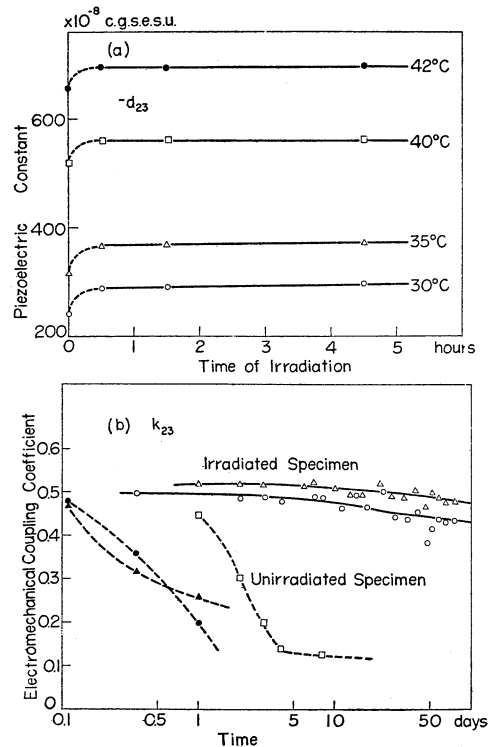


Fig. 9. Irradiation Effects by X-ray.

against depolarization, especially near the Curie point. Fig. 9(b) shows some of the aging data at 42°C as for many specimens, where the electromechanical coupling coefficients k_{23} are plotted against time in days. In all specimens prepoled but unirradiated, values of k_{23} rapidly decrease, while irradiated samples are found to be considerably stable. Some data in the case of γ -ray irradiation are shown in Fig. 10(a) and (b):

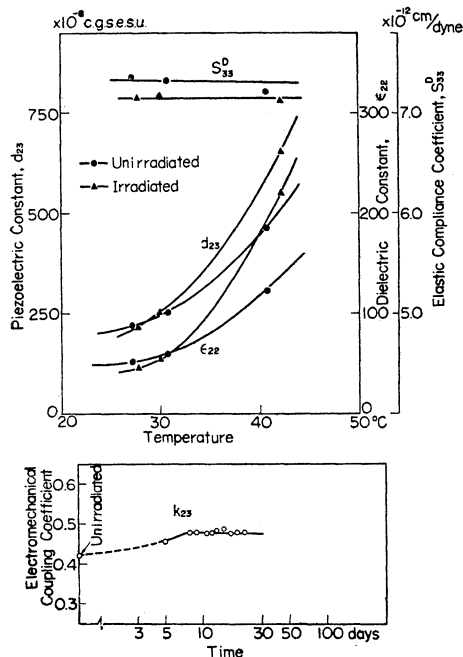


Fig. 10. Irradiation Effects by γ -ray.

Temperature dependences of some constants are shown in (a) and an example of aging data is shown in (b). Similar results as above are obtained in these cases.

§ 5. Conclusion

The piezoelectric and electrostrictive properties of triglycine sulphate crystal are studied here from a stand point of an "induced" piezoelectricity. We obtain the following electrostrictive constants:

$$Q_{21}=1.5, \quad Q_{22}=2.5, \quad Q_{23}=-4.4,$$

$$Q_{25}=0.4, \quad Q_{44}=-4.3, \quad Q_{66}=2.2$$

and $Q_{46}=4.2$

in 10^{-11} c.g.s. units.

The first four constants are found to be

nearly independent of temperature, but as for the remaining ones, it is not confirmed whether they remain constant or not against temperature, because of large influences of errors in determining d_{2i} 's.

Some elastic compliance coefficients s^{D} 's for constant electric displacement are found to show no anomalies at the Curie point, and it will be concluded that the phase transition at 47.5°C is not accompanied by any elastic change.

Even in a prepoled crystal, when it is once heated above the Curie point and cooled again, many antiparallel domains grow and a piezoelectric constant decreases apparently. If the crystal is irradiated by x-ray or γ -ray under a high electric fields, it is found that such a decrease is prevented without biasing field. This fact will give a possibility to utilize triglycine sulphate crystals as piezoelectric devices.

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